

Question	Marking Guidance	Mark	Comments
06.1	A. Cell body OR Soma OR Dendron/dendrite(s) B. Axon OR Axoplasm C. Node of Ranvier D. Schwann cell;;	2 (2 x AO1)	4 correct = 2 marks 2–3 correct = 1 mark 0–1 correct = 0 marks For B ignore cytoplasm For D accept shwan(n), scwan(n), scwaan and phonetic spellings of schwann but reject 'swan'

Question	Marking Guidance	Mark	Comments
06.2	1. No/less saltatory conduction OR Depolarisation/impulses doesn't occur from node to node OR Depolarisation/impulses/ions pass to other neurones OR Depolarisation/impulses occurs along whole length (of axon); 2. (Nerve) impulses slowed; 3. (Affects/involves) relay neurones (in brain) OR (Affects transmission) to motor neurone/s OR (Involves) muscle/effector/neuromuscular junction;	3 (3 x AO2)	1 and 2. Ignore 'messages' or 'signals' 1 and 2. Accept 'action potential(s)' for 'depolarisation/impulses' 2. Accept 'impulse(s) take longer to travel' Mark points 1 and 3 can still be awarded if answer indicates nerve impulses travel faster

Question	Marking Guidance	Mark	Comments
06.3	1. Less/no acetylcholine binds (to receptors); 2. Fewer/no impulse(s)/action potentials (at synapse before SAN) OR Less/no depolarisation by PNS (at synapse before SAN) OR PNS/acetylcholine decreases heart rate OR More depolarisation occurs (as acetylcholine is inhibitory at SAN) OR Less/no inhibition by PNS/acetylcholine (at SAN); 3. (Refers to) SAN;	3 (3 x AO1)	1 and 2. Accept neurotransmitter or abbreviations for e.g. ACh, Ach (as named in question) for 'acetylcholine' 1. Ignore competitive inhibition unless it refers to enzyme or active site, then reject 2. Note: First two options relate to atropine acting at synapse in PNS before SAN. Last 2 options relate to atropine acting directly on SAN 2. Ignore 'messages' or 'signals'

Question	Marking Guidance	Mark	Comments
06.4	Correct answer of 0.687 = 2 marks;; Answer shows 687 (ignore position of decimal point and preceding / subsequent zeros i.e. incorrect order of magnitude) = 1 mark OR Answer of 0.687 (to more than 3 significant figures e.g. 0.6874 / 0.68737) = 1 mark OR Answer of 0.69 / 0.7 (i.e. to fewer than 3 significant figures) = 1 mark OR Answer of 0.688 (incorrect rounding) = 1 mark OR	2 (2 x AO2)	

	Answer of 1.54 (added rather than multiplied by BMI) = 1 mark OR Answer of 1.81×10^{-4} (multiplied by (216 x BMI) rather than added) = 1 mark ;		
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